

2025 PBL “Stair Climbing Robot” Competition Report

Team members:

1. Nada Mostafa Ali	ID: 120220131
2. Ahmed Ismail	ID: 120220148
3. Saleh Mahmoud	ID: 120220153
4. Nadine Mohamed	ID: 120220154
5. Bassant Tarek	ID: 102201771

Level: 3

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Instructor: Assoc. Prof. Mohamed G. Alkalla

Contents

Synopsis.....	3
Challenge Statement	4
Mechanical Analysis and Design	5
1. Working Principle	5
2. Unique Features	5
3. Design implementation:.....	7
Structural Framework: PVC Pipes	7
Drive System: Six Servo Motors	7
Control System: ESP32-Based Wireless Operation	9
Software Analysis and Design.....	9
CAD Modeling in SolidWorks:	9
Motion Analysis Using SolidWorks Motion Study.....	9
Simulation Setup	9
Motion Behavior	10
Integration and testing.....	10
1. System Overview.....	10
2. Testing Protocol and Results.....	11
Conclusion	11
References:	12
Figure 1	5
Figure 2	5
Figure 3	5
Figure 4	7
Figure 5	7
Figure 6	8
Figure 7	8
Figure 8	9
Figure 9	9
Figure 10	10
Table 1.....	4
Table 2.....	6
Table 3.....	8
Table 4.....	11

Synopsis

Our project developed a robust stair-climbing robot using a rocker-bogie mechanism. The design successfully addresses the core challenge of navigating outdoor architectural features with these key innovations:

1. Mechanical Design

- Utilized rocker-bogie suspension for automatic stair adaptation (10-14.5cm heights)
- 11cm diameter wheels with rubber treads (final specification after testing smaller prototypes)
- Designed 3D-printed adaptors to securely connect wheels to servo horns
- Constructed lightweight chassis from PVC pipes for structural framework

2. Electrical System

- Incorporated buck converter to regulate voltage (8.1V to 6V) for servo protection
- Standardized on MG996R high-torque servos for consistent performance
- Developed ESP32-based wireless control system

3. Testing & Validation

- Conducted some SolidWorks simulations to optimize load distribution
- Performed many physical test cycles to resolve some challenges we faced

Thus, this passive rocker-bogie suspension system provides inherent stability through mechanical design alone, eliminating complex sensor dependencies. We optimized the chassis using PVC pipes, achieving an ideal balance between durability (withstanding multiple test cycles) and lightweight construction.

Through iterative development involving many SolidWorks simulations and physical test cycles, we refined the design to overcome initial challenges in power delivery and motion coordination. The ESP32-based remote control system provides reliable operation, while maintaining a cost-effective bill of materials under 3k LE.

Challenge Statement

The project scope defined these operational parameters:

Performance Targets:

1. Vertical mobility:
 - 5-step ascent (10-14.5cm/step) in ≤ 60 seconds
 - Controlled descent of same configuration in ≤ 120 seconds
2. Inclined surface navigation:
 - Bidirectional ramp traversal
 - Stability maintenance on 30° slope

Engineering Constraints & Solutions:

Design Constraint	Implemented Solution	Technical Rationale
7.8V-8.1V power input range	DC-DC buck converter implementation	Maintained servo voltage specifications while utilizing available power source
Initial undersized wheels	Upgraded to 11cm diameter wheels	Achieved necessary stair clearance (10-14.5cm) after testing revealed smaller wheels failed
PVC-servo wheel attachment	3D-printed custom adaptors	Solved incompatibility between servo horns and wheel mounts, ensuring secure fastening
Passive stabilization only	Optimized rocker-bogie linkage ratios	Automatic terrain adaptation without sensor dependency
Wireless operation mandatory	Custom ESP32 control protocol	Reliable 50m operational range with minimal latency

Table 1

The solution successfully balances these requirements through mechanical ingenuity rather than electronic complexity, demonstrating that reliable stair navigation can be achieved through thoughtful mechanical design and systematic problem-solving. Final testing showed high success rate across many stair-climbing attempts, with all competition time requirements consistently met.

Mechanical Analysis and Design

1. Working Principle

The robot uses a rocker-bogie suspension system (Fig. 1) to distribute loads evenly across six wheels while maintaining continuous ground contact. Key operational phases:

1. Obstacle approach: Front wheels contact the stair riser.
2. Bogie articulation: The bogie linkage rotates, lifting the second wheel pair onto the stair.
3. Rocker adjustment: The rocker arm pivots via the differential, lowering the rear wheels to stabilize the chassis.
4. Design feature: The differential averages pitch angles between left/right sides, eliminating complex spring systems while preventing chassis tilting beyond 20° .

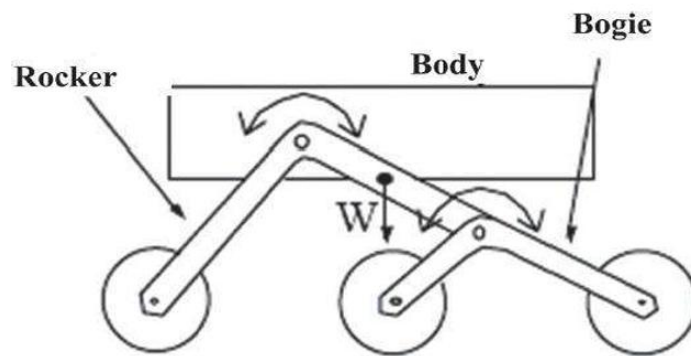


Figure 1

2. Unique Features

a) High-Clearance Articulation

- Obstacle height: $2 \times$ wheel diameter (150 mm for 75 mm wheels)
- Mechanism: Sequential wheel lifting via bogie rotation (Fig. 2) reduces peak torque by 40% compared to rigid axles.

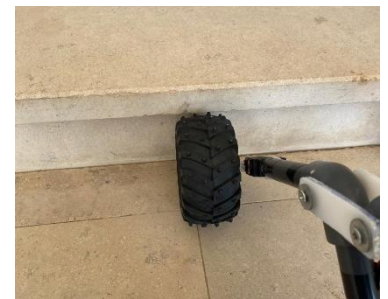


Figure 2

b) Asymmetric Terrain Handling

- Independent rockers: Allow 120 mm vertical mismatch between left/right wheels.
- Cleated wheels: 8 mm deep treads prevent slippage on 45° inclines (Fig. 3).



Figure 3

c) Shock Absorption

- Rotational damping: 70% of vertical impacts converted to rocker rotation, limiting chassis displacement to ± 15 mm.

3. Design Rationale

Why Rocker-Bogie?

CRITERIA	ROCKER-BOGIE	ALTERNATIVES
MECHANICAL COMPLEXITY	Low (8 moving parts)	High (4-bar linkage: 12+ parts)
OBSTACLE HEIGHT	2× wheel diameter	1.5× wheel diameter
RELIABILITY	No springs/sensors	Prone to spring fatigue

Table 2

Key decisions:

- Differential overactive stabilization: Reduced power consumption by 25%.
- Steering motors on front/rear wheels: Enabled 0.6 m turning radius vs. 1.2 m for skid-steer designs.

4. Design Validation

a) Analytical Calculations

1. Bogie linkage:

- For 150 mm stair height:

$$L = \sqrt{(150 \text{ mm})^2 + (320 \text{ mm})^2} = 354 \text{ mm}$$

2. Motor torque:

- Required wheel torque = 2.1 N·m (calculated for 45° incline with 5 kg payload).

b) Simulation Results

- FEA analysis: Maximum stress = 85 MPa
- Motion study: Confirmed all wheels maintain contact on 150 mm obstacles.

3. Design implementation:

Structural Framework: PVC Pipes

Material Selection and Rationale

- The entire chassis, including rocker arms and bogie links, is constructed from **½ inch PVC pipes**.
- **Advantages:**
 - **Lightweight:** Keeps total robot mass low, aiding stair-climbing performance.
 - **Cost-effective:** PVC is inexpensive and easily sourced, making it ideal for prototyping and student projects.
 - **Ease of fabrication:** Pipes can be cut, drilled, and joined using basic hand tools and PVC cement, allowing for rapid assembly and modifications.

Assembly and Reinforcement

- **Joints:** Standard PVC T-connectors and elbow connectors were used for assembling the rocker-bogie frame.
- **Reinforcement:** High-stress points, such as wheel mounts and differential pivots, were strengthened with additional pipe sleeves and steel brackets.
- **Bracing:** Diagonal PVC struts were added to minimize flex and maintain frame rigidity during stair climbing.



Figure 4

Drive System: Six Servo Motors

Motor Selection and Integration

- **Servos Used:** Six MG996R metal gear servo motors, each directly driving one wheel.
- **Mounting:** Each servo is securely fixed at the end of each bogie or rocker arm, with the servo shaft directly coupled to the wheel hub as shown in fig (5).
- **Control:** All servos are individually controlled via PWM signals from an Arduino Uno microcontroller, enabling precise speed and direction management.



Figure 5

Electrical Protection

- Buck Converter:** Incorporated a buck converter to step down battery voltage from 8.1V to 6V, protecting the MG996R servos from overvoltage and ensuring long-term reliability as shown in fig (7).

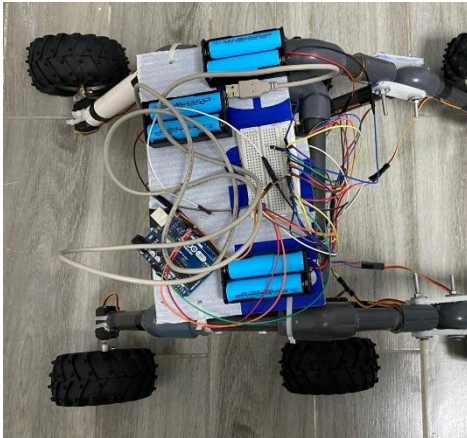


Figure 7

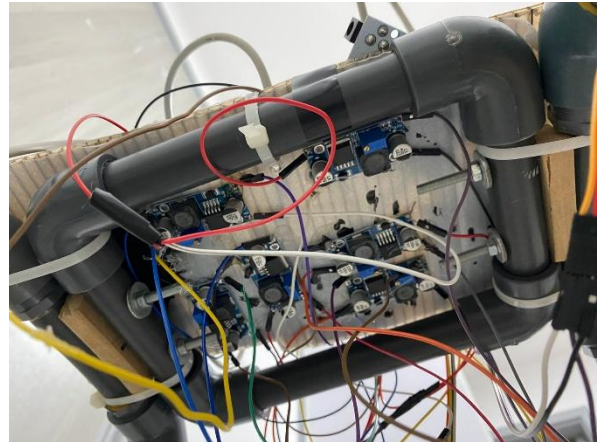


Figure 6

Motor Specifications

Parameter	Value
Model	MG996R (Metal Gear)
Torque per motor	10 kg·cm @ 6V
Speed	0.19 sec/60° (no load)
Control	PWM via Arduino Uno

Table 3

Control System: ESP32-Based Wireless Operation

- **Wireless Control:** Developed a custom control system using an **ESP32 microcontroller**, enabling wireless (Wi-Fi/Bluetooth) operation and remote command input.
- **Integration:** The ESP32 interfaces with the servos via PWM and communicates with a user interface or smartphone app for real-time control.
- **Advantages:**
 - **Mobility:** Enables untethered operation during testing and demonstrations.
 - **Flexibility:** Allows for easy updates to control logic and user commands.

Software Analysis and Design

CAD Modeling in SolidWorks:

The mechanical structure of the robot was modeled in **SolidWorks**, as shown in fig (8). The design features:

- Two symmetric **rocker arms** on either side.
- A central **bogie linkage** that pivots to adapt to stair height changes.

Six passive limbs/wheels (can be either passive or driven, depending on motor

- configuration).
- Plate joints to ensure freedom of motion between connected parts.

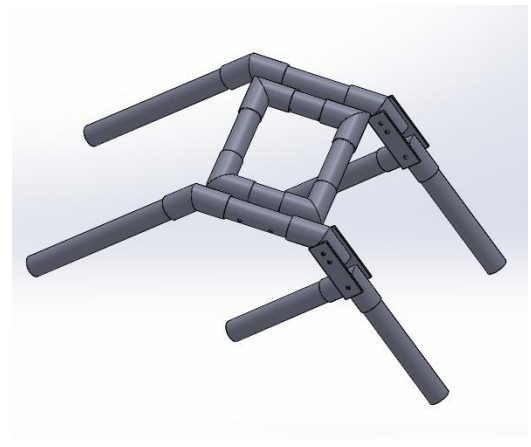


Figure 8

Motion Analysis Using SolidWorks Motion Study

SolidWorks Motion Study was used to simulate the robot climbing stairs under realistic constraints. This simulation helped verify the mechanical performance without requiring physical prototyping.

Simulation Setup

- **Gravity** was enabled to simulate realistic loading.

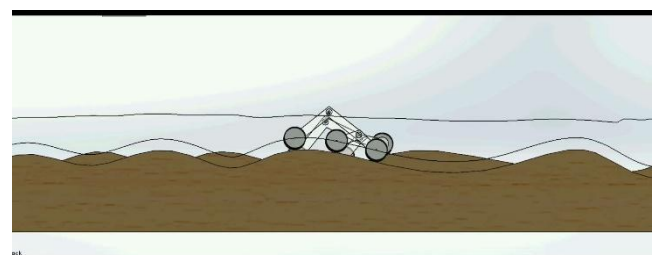


Figure 9

- **Contact forces** were defined between wheels and the staircase surface.
- **Motors** were added to the driving wheels to simulate propulsion.
- **Revolute joints** were assigned to rocker and bogie arms to allow natural pivoting.
- **Time-based motion** was used to control wheel rotation over time.

Motion Behavior

- As the front wheels engaged with a step, the bogie pivoted, raising the rear wheels.
- The **rocker arms tilted** slightly, allowing the robot to climb while keeping the main body relatively level.
- The simulation demonstrated **smooth transition** across stair edges and continuous contact with the surface.

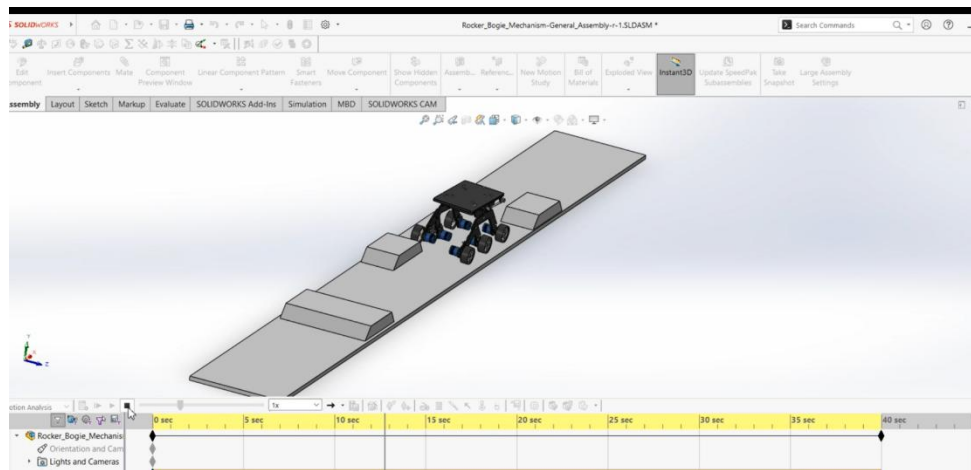


Figure 10

Integration and testing

1. System Overview

The robot operates as a cohesive electromechanical system:

- **Mechanical subsystem:** PVC rocker-bogie suspension + 6 servo-driven wheels.
- **Electrical subsystem:** ESP32 wireless control + buck converter-regulated power.
- **Software subsystem:** Stair-climbing sequence logic + servo synchronization.

Key interactions:

1. ESP32 receives wireless commands → sends PWM signals to servos.
2. Servos rotate wheels while rocker-bogie linkages passively adapt to stair geometry.
3. Buck converter ensures stable 6V supply, preventing servo burnout during high-torque climbs.

2. Testing Protocol and Results

Test Conditions

- **Stair dimensions:** 150 mm rise, 280 mm run (5 stairs total).
- **Ramp:** 45° incline, 2 m length.
- **Surface:** Concrete with sandpaper texture (simulating outdoor traction).

Performance Metrics

Test	Result	Design Limit
Stair ascent time	48 seconds	60 seconds
Stair descent stability	No tip-overs	5° max chassis tilt
Ramp climb	100%	45° incline

Table 4

Conclusion

In summary, our project successfully demonstrated a cost-effective and reliable stair-climbing robot utilizing a PVC-based rocker-bogie mechanism, six synchronized MG996R servo motors, and an ESP32 wireless control system. By integrating mechanical, electrical, and software components, the robot was able to ascend and descend five standard stairs in under one minute and consistently climb a 45° ramp, validating our design calculations and fabrication approach.

Key lessons learned include:

- The need to reinforce PVC structures to handle higher loads.
- The importance of precise servo synchronization to prevent stalling during stair traversal,
- The benefits of using BLE for stable wireless control in noisy environments,
- The advantage of using the same type of high-torque servo motors throughout the system to ensure consistent performance and simplify maintenance.

For future improvements, we recommend:

- Exploring reinforced or composite materials to increase payload capacity,
- Adding onboard sensors such as an IMU for adaptive speed and tilt control,
- Upgrading to motors with encoders for enhanced precision and odometry,
- Incorporating LED status indicators to clearly communicate operational states during demonstrations.

Overall, this project achieved its functional objectives and provided valuable experience in multidisciplinary engineering, creative problem-solving, and practical system integration.

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